

Hematology Patient Population Analysis: SQL Deep Dive (Synthea Synthetic Data)

Project Overview

This project presents an end-to-end SQL analysis of a synthetic patient population generated by Synthea, an open-source synthetic patient simulator (MITRE). The objective was to identify the hematology-related patient subpopulation within the dataset and apply progressively advanced SQL techniques to uncover clinically meaningful patterns in diagnosis, treatment, and healthcare utilization.

Objectives

- Identify and isolate the hematology-related patient population.
- Clean and prepare imported data for analysis.
- Apply SQL techniques of increasing complexity: filtering, aggregation, CTEs, and window functions.

- Cross-reference conditions, medications, and encounters to build a clinically coherent interpretation.

Dataset

- Study population: Synthetic patients generated by Synthea (~1,000 patients).
- Sample size: 300 patients with a diagnosis of Anemia.
- Variables: Age, sex, diagnosis, encounter type and frequency, medications, and encounter dates.
- All data is synthetic and anonymized by design; no real patient information is included.

Tools and Technologies

SQL Server Management Studio (SSMS), T-SQL

Key Findings

- Female patients: 135, average age 61. Male patients: 165, average age 72.

Demographics by Sex

- Epoetin Alfa was the dominant anemia treatment (3,388 prescriptions) compared to oral iron (30

prescriptions), suggesting the majority of cases are renal-related rather than nutritional.

Treatment Distribution

- Patients with the highest encounter frequency (up to 2,006 encounters) were predominantly ambulatory (73%), a pattern consistent with recurring dialysis treatment – reinforcing the renal-anemia interpretation drawn from the medication data.

Encounter Types

Skills Demonstrated

Clinical data cleaning, Exploratory Data Analysis (EDA), Common Table Expressions (CTEs), window functions (RANK() OVER), multi-table joins, clinical interpretation of data, healthcare data analytics using SQL.